NGINX Deployment

Movie Collection API

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Deployment setup

The local deployment uses:

- MySQL for movies and genres.
- PHP-FPM to run the API code.
- NGINX to receive HTTP requests on port 8081.

NGINX forwards the path, body, method, and Authorization header.
The API handles the request and returns JSON.

Install and prepare

Ubuntu 24.04 example:

```
sudo apt install nginx php-fpm php-mysql mysql-server
sudo systemctl start mysql
sudo mysql < code/schema.sql
sudo mkdir -p /var/www/movie-api/code
sudo cp code/* /var/www/movie-api/code/
```

Create the database user using README.md.

Do not use the MySQL root account for the API.

NGINX server configuration

Save `code/nginx.conf.example` as `/etc/nginx/sites-available/movie-api`.

```
server {
    listen 127.0.0.1:8081;
    server_name localhost;
    root /var/www/movie-api/code;
    # Locations are shown on the next slide.
}
```

The root path must match the location of the PHP files.

Routing to PHP-FPM

Inside the server block:

```
location = /test_client.html { default_type text/html; }
location / {
    include fastcgi_params;
    fastcgi_param SCRIPT_FILENAME $document_root/index.php;
    fastcgi_param HTTP_AUTHORIZATION $http_authorization;
    fastcgi_pass unix:/run/php/php8.3-fpm.sock;
}
```

Only the test client is served as a static file.

Source: NGINX FastCGI documentation, linked in README.md.

Configure PHP-FPM

Edit `/etc/php/8.3/fpm/pool.d/www.conf`.

Add these under the existing `[www]` pool:

```
env[DB_HOST] = 127.0.0.1
env[DB_PORT] = 3306
env[DB_USER] = movie_app
env[DB_PASSWORD] = YOUR_LOCAL_PASSWORD
env[API_TOKEN] = YOUR_LOCAL_TOKEN
```

Use your own local values. Keep them outside GitHub.

Adjust the PHP version in the socket and service if needed.

Enable and start

```
sudo ln -s /etc/nginx/sites-available/movie-api \
  /etc/nginx/sites-enabled/movie-api
sudo php-fpm8.3 -t
sudo nginx -t
sudo systemctl restart php8.3-fpm nginx
curl -i http://localhost:8081/movies
```

A working response has status 200 and movie JSON.

Check that the HTTP Server header identifies NGINX.

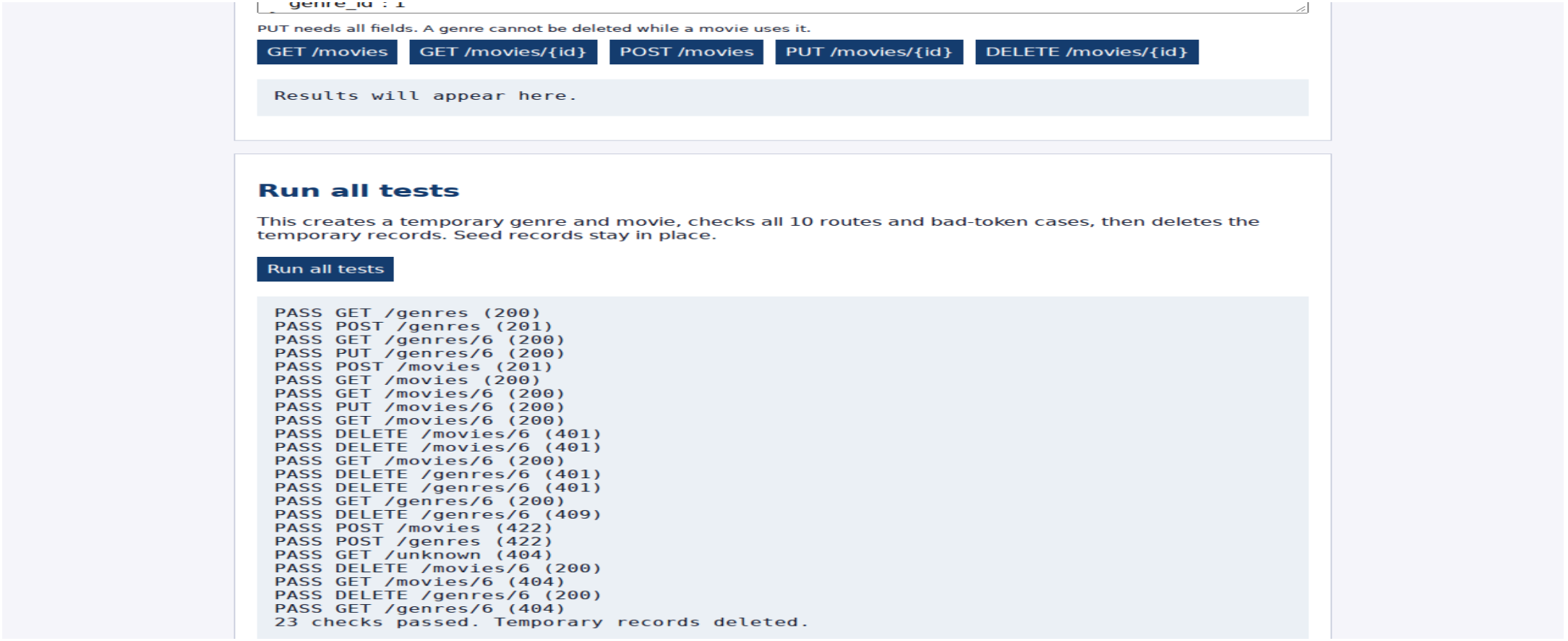
Working API capture



Real GET /movies result from the browser client through NGINX on port 8081.

The response header was also checked for `nginx/1.24.0` .

Browser test capture



The same browser suite passed through NGINX.

Test results

```
BASE_URL=http://localhost:8081 bash code/test_curl.sh
```

- cURL: 29 checks passed through NGINX.
- Browser: 23 checks passed through NGINX.
- Missing and wrong tokens returned 401 on both delete routes.
- A genre in use returned 409.
- Deleted records returned 404 on later reads.

See `verification.md` for the exact tested environment.

Troubleshooting and stopping

502: Check PHP-FPM is running and its socket matches NGINX.

500: Check the database user, password, and schema import.

401: Check the request token matches PHP-FPM's API_TOKEN.
Restart FPM after changing pool settings.

```
sudo systemctl stop nginx php8.3-fpm
```

The deployment is local. It is not a public hosted website.

Sources

- Supplied Project 1 overview, deployment requirements.
- NGINX: Module `ngx_http_fastcgi_module`. https://nginx.org/en/docs/http/ngx_http_fastcgi_module.html
- PHP: Runtime Configuration (PHP-FPM). https://www.php.net/manual/en/install.fpm.configuration.php
- MySQL 8.0 Reference Manual: FOREIGN KEY Constraints. https://dev.mysql.com/doc/refman/8.0/en/create-table-foreign-keys.html

Screenshots are from the working local project.